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DESIGN AND IMPLEMENTATION OF SECURITY INFORMATION AND EVENT MANAGEMENT(SIEM) SYSTEM FOR MONITORING SECURITY EVENTS IN A BUSINESS ORGANIZATION

PRESENTED BY

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ABSTRACT

In today's digital era, business organizations face increasing cybersecurity threats such as malware attacks, phishing, insider threats, and unauthorized access. Managing and monitoring security events manually is inefficient and prone to errors due to the large volume of logs generated across an organization's IT infrastructure. To address this challenge, the proposed system implements a **Security Information and Event Management (SIEM)** solution that collects, analyzes, and monitors security logs in real time. The system aggregates logs from various sources including servers, firewalls, routers, endpoints, and applications, and correlates these events using predefined security rules to identify suspicious activities. When potential threats are detected, the system generates real-time alerts that help administrators respond quickly to security incidents. This project demonstrates how implementing a SIEM system enhances an organization's cybersecurity posture by enabling proactive monitoring, efficient log management, and intelligent threat detection.

INTRODUCTON

- Cybersecurity threats have increased significantly due to the growing dependence on digital infrastructure. Organizations face multiple threats including malware attacks, phishing, insider threats, data breaches, and unauthorized access.
- Security Information and Event Management (SIEM) systems provide a centralized platform that collects logs from multiple sources and analyzes them to identify security incidents.
- The main objective of this project is to design and implement a SIEM system that helps organizations monitor security events effectively and detect potential threats in real time.
- Functions are

  a) Log Collection b) Event Correlation and analysis
  c) Threat Detection d) Real-time monitoring
  e) Security alert generation f) Compliance reporting

LITERATURE REVIEW

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Tite	Author(s)	Year	Algorithms Used
Security Information and Event Management (SIEM): A Review	Behl & Behl	2021	Overview of SIEM architecture and log correlation techniques
Log Management and Security Monitoring using SIEM	Kent & Souppaya	2020	Discusses importance of centralized log monitoring
Security Event Correlation Techniques	Julisch	2019	Event correlation for intrusion detection
Practical SIEM Implementation	Shackleford	2022	Methods for deploying SIEM in enterprise environments

EXISTING SYSTEM

Many organizations currently rely on traditional security monitoring systems such as firewalls, antivirus software, and manual log analysis.

Limitations of existing systems include:

a) Logs are stored separately across different systems

b) Manual Monitoring

c) Difficult to detect advanced cyber attacks

d) Lack of Centralized monitoring

e) Delayed incident Detection

PROPOSED SYSTEM

The proposed system implements a **Security Information and Event Management (SIEM)** solution to monitor and analyze security organization's IT infrastructure.

Key features include:

a) Centralized log collection from multiple devices

b) Real-time event monitoring

c) Event correlation to detect suspicious activities

d) Security alert generation

e) Dashboard for monitoring security events

f)Log storage and analysis

MODULE DESCRIPTION

Project Modules

- Log Collection Module
- Log Processing Module
- Event correlation module
- Alert Generation Module
- Monitoring Dashboard
- Reporting Module

MODULE DESCRIPTION

Data Collection and Processing

The collected logs may include information such as:

a) User Login activities
b) Network Traffic events
c) System Errors and warnings
d) Firewall alerts
e) Application Access logs

Log processing steps include:

a) Filtering irrelevant logs
b) Parsing log messages
c) Normalizing log formats
d) Strong logs in a centralized log database

MODULE DESCRIPTION

Event Correlation and Analysis

The system analyzes logs using predefined security rules such as:

a) Multiple failed login attempts

b) Access from unknown IP addresses

c) Unauthorized privilege escalation

d) Malware activity activity detection

e) Abnormal network traffic patterns

MODULE DESCRIPTION

Security Model Evaluation

- The system classifies events into categories such as:

  a) Information Events b) Warning Events c) Critical Incidents

- Evaluation Criteria Include:

  a) Frequency of Suspicious events b) Type of activity detected

  c) Source of the event d) Risk level of affected system

This evaluation helps administrators prioritize responses and quickly handle high-risk threats.

MODULE DESCRIPTION

Security Monitoring and Alert System

- The SIEM system continuously monitors incoming logs and identifies potential security threats in real time.

  a) Generates alerts b) Sends notification to administrators
  c) Displays warnings on the monitoring dashboard

- Examples of alerts include:

  a) Unauthorized login attempts b) Malware detection
  c) Firewall policy violations d) Suspicious network activity

MODULE DESCRIPTION

Database Integration

- The SIEM system integrates a centralized database to store all collected security logs and detected events.
- Database stores information such as:
  a) Log records b) Security Alerts c) Incident reports
  d) User Activity logs

This Centralized storage allows administrators to:
  a) Search historical logs b) Perform forensic analysis
  c) Generate security reports d) Track previous incidents

Databases commonly used in SIEM systems include
  a) Elasticsearch b) MySQL c) MongoDB

MODULE DESCRIPTION

User Interface

- The user interface provides a dashboard that allows administrators to monitor security events in real time.

The User displays:

a) Security alerts

b) System status

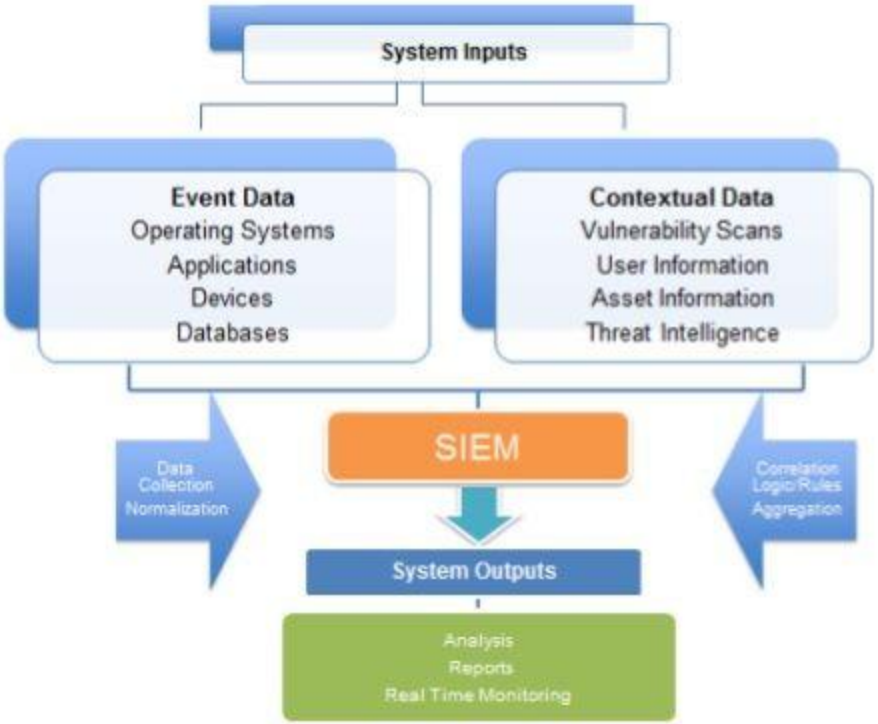
c) Log Statistics

d) Threat detection reports

e) Event Timelines

ARCHITECTURE

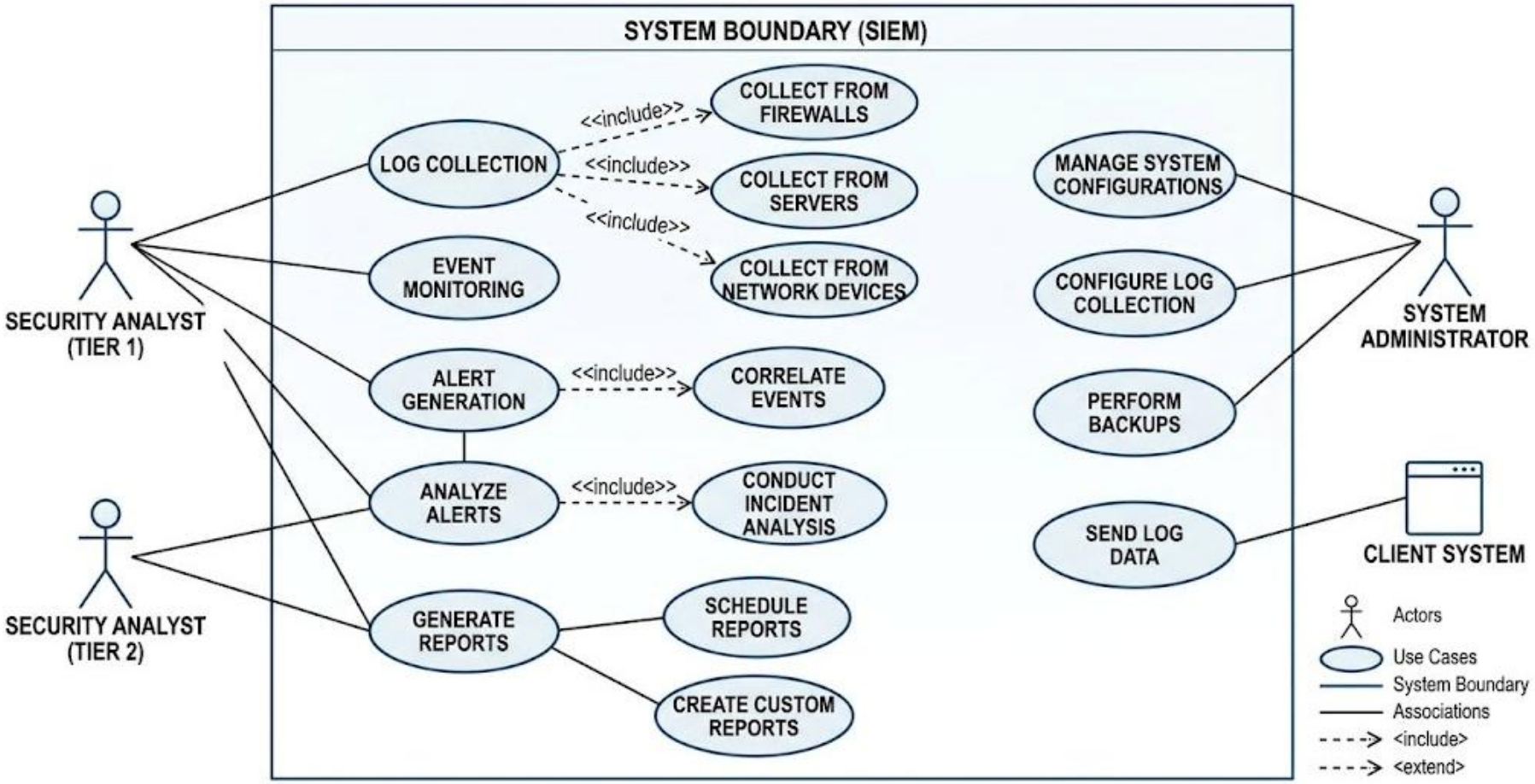
SIEM Architecture



UML DIAGRAMS

- UML represents Unified Modelling Language.
- UML diagram is a way to visualize systems and software using Unified Modelling Language (UML).
- It is a standardized modelling language used in software engineering to visualize, specify, construct, and document the artifacts of software systems.
- Some of those are:-
  - ❑ Use Case Diagram
  - ❑ Class Diagram
  - ❑ Sequence Diagram
  - ❑ Activity Diagram
  - ❑ Component Diagram
  - ❑ Deployment Diagram

USE CASE DIAGRAM

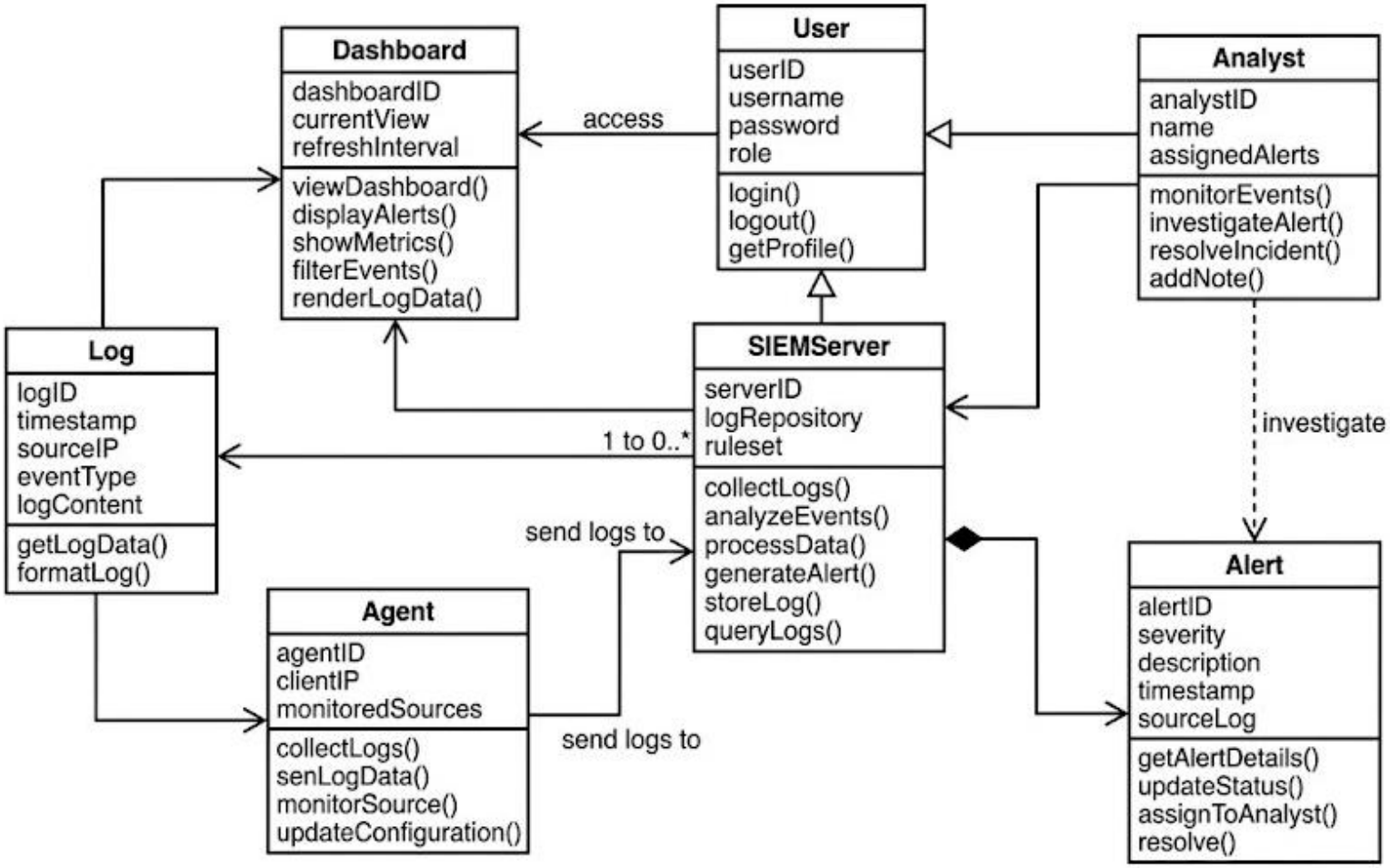


USE CASE DIAGRAM

- **Core Functionality:** It shows that "Log Collection" is the foundation, which includes gathering data from firewalls, servers, and network devices.

- **Analysis Workflow:** Analysts focus on monitoring events, analyzing alerts, and generating reports, while the Administrator handles the "behind-the-scenes" work like backups and configurations.

CLASS DIAGRAM

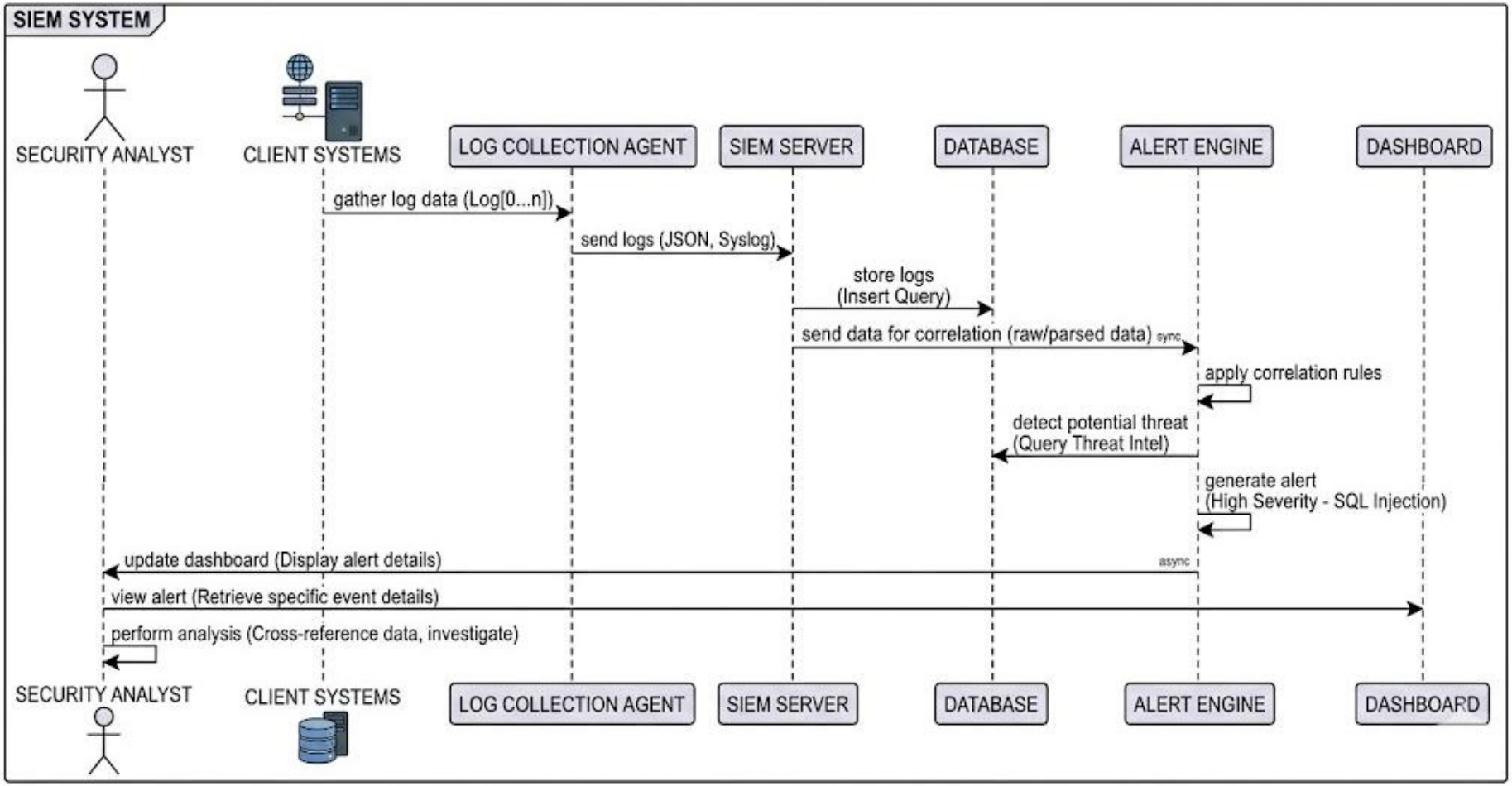


CLASS DIAGRAM

- **Data Relationships:** It defines the Log and Alert classes as the primary data types. A Log is what comes in, and an Alert is what is produced if something looks suspicious.

- **System Logic:** The SIEMServer is the "brain," containing methods like analyzeEvents() and generateAlert(). It connects the Agent (the collector) to the Dashboard (the interface).

SEQUENCE DIAGRAM



SEQUENCE DIAGRAM

- **The Lifecycle of a Threat:** It shows the journey of data: starting from "Gather log data" (Client Systems) "Send logs" (Agent) "Store & Correlate" (Server/Database) "Generate Alert.“

- **Human Interaction:** It specifically highlights how a "High Severity" alert (like an SQL Injection) triggers an update to the Dashboard, which the Security Analyst then investigates.

ACTIVITY DIAGRAM

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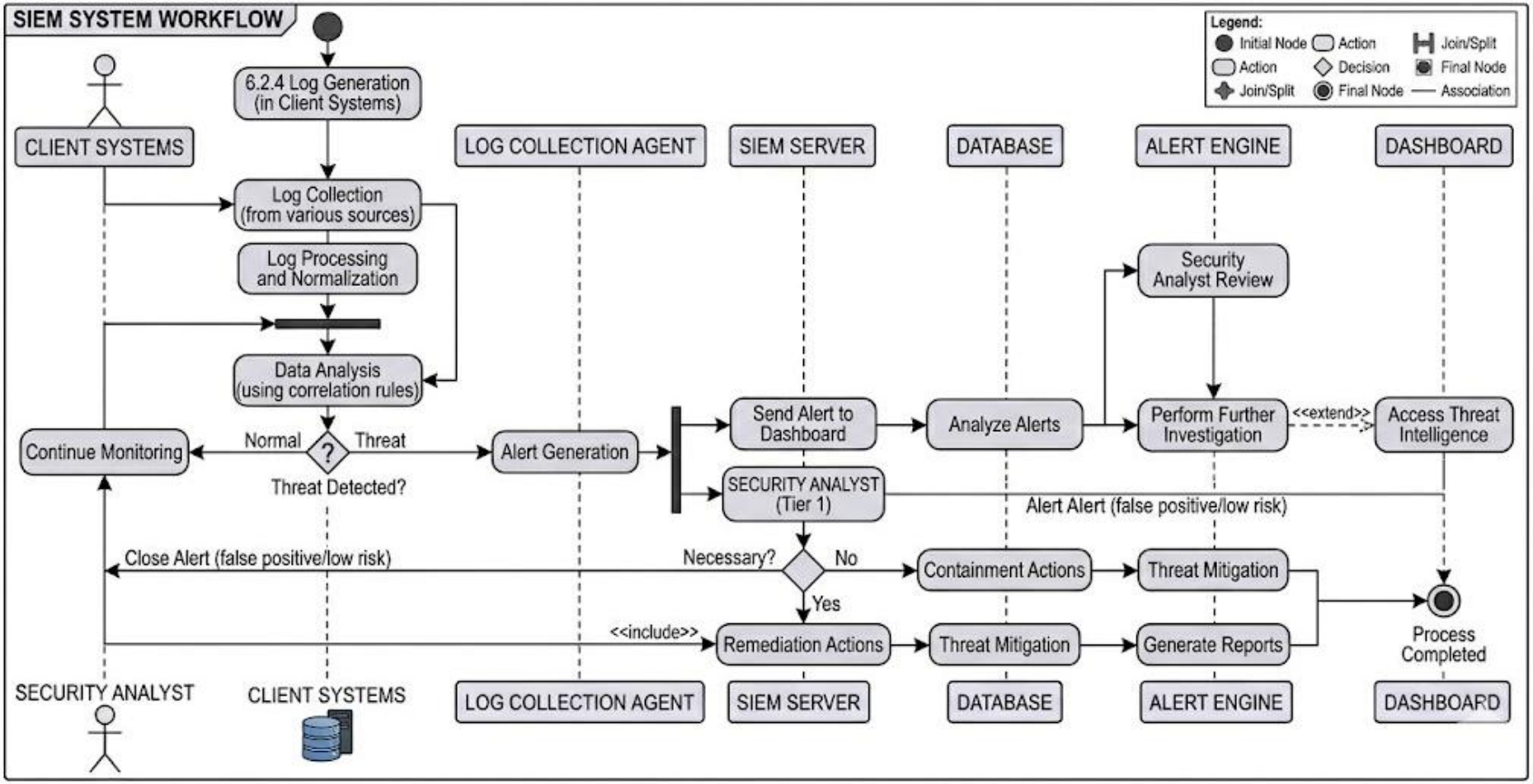
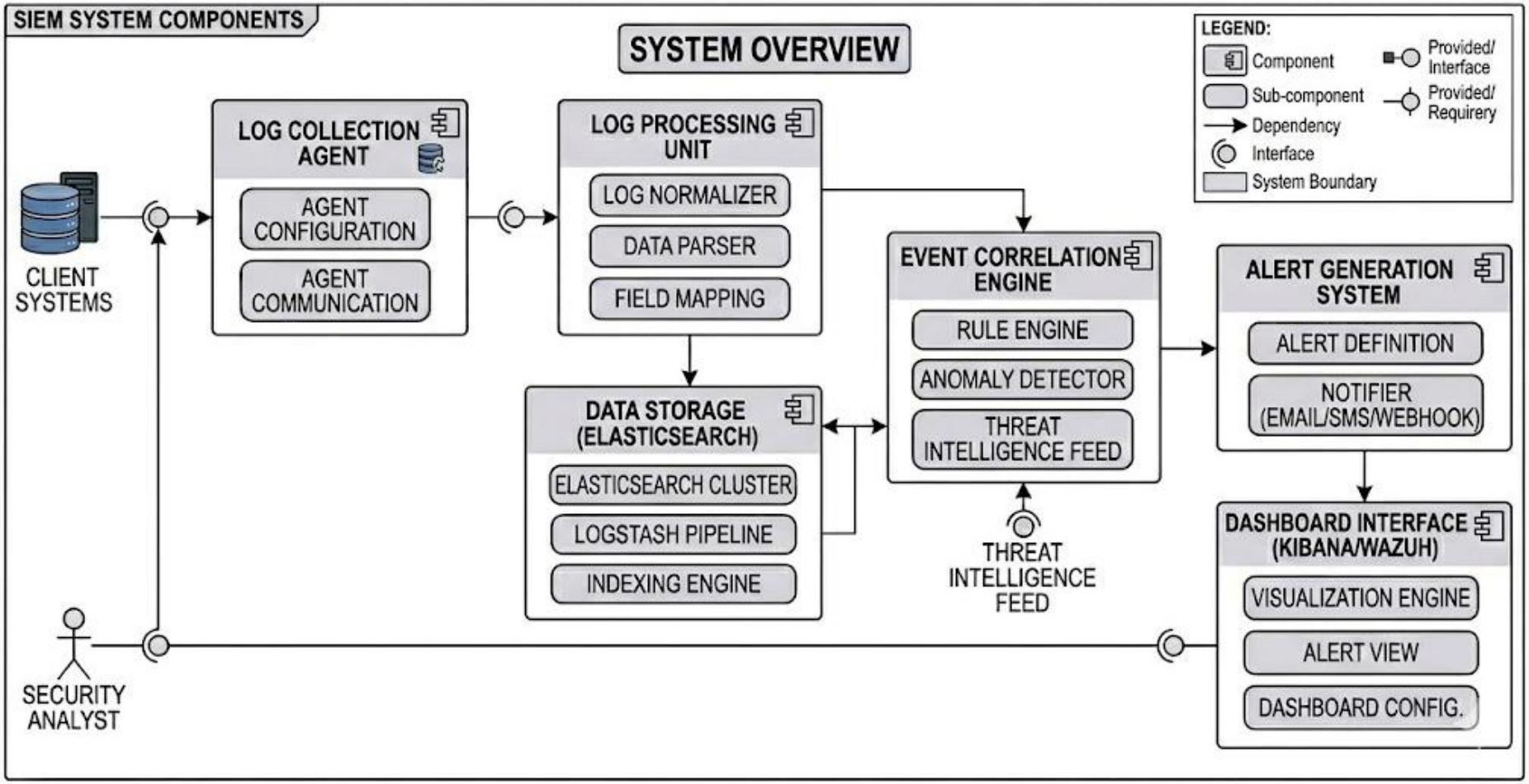


ACTIVITY DIAGRAM

- **Decision Points:** The most critical part is the diamond-shaped "Threat Detected?" node. If "No," it continues monitoring; if "Yes," it triggers the Alert Generation process.

- **Remediation:** It maps out the "Investigation" phase where a Tier 1 Analyst decides if containment or remediation actions are necessary to mitigate a threat.

COMPONENT DIAGRAM

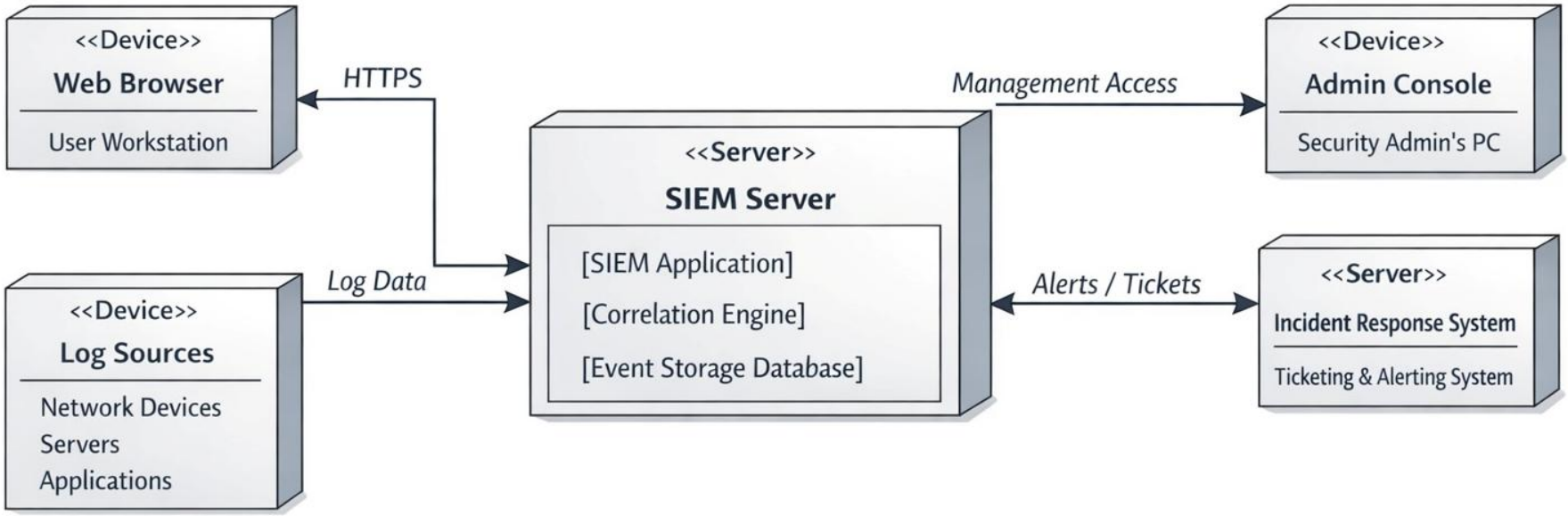


COMPONENT DIAGRAM

- **Modular Design:** The system is broken into distinct units like the "Log Processing Unit" (which cleans data) and the "Event Correlation Engine" (which looks for patterns).

- **External Integration:** It highlights the use of Elasticsearch for storage and Kibana/Wazuh for the dashboard interface, showing how the SIEM leverages existing powerful database and visualization tools.

DEPLOYEMENT DIAGRAM



DEPLOYEMENT DIAGRAM

- **Core Infrastructure:** The SIEM Server is the central hardware node acting as the primary host for the SIEM Application, Correlation Engine, and Event Storage Database. It serves as the processing hub for all incoming security data.

- **Data Flow and Access:** Log Data is ingested from external sources (Network Devices, Servers, and Applications), while the system provides visibility through a Web Browser via HTTPS and allows for administrative control via the Admin Console.

- **Incident Integration:** The system connects directly to an Incident Response System to exchange Alerts and Tickets, ensuring that detected threats are immediately moved into a ticketing workflow for remediation.

SYSTEM REQUIREMENTS

Software Requirements:-

Application platform	: Python 3.0 & above
Operating System	: Linux / Windows
Database	: Elastic Search
IDE	: Visual Studio Code

Hardware Requirements:-

RAM	: 8 GB & Above
Processor	: Intel I3 & Above
CPU	: 1.6GHz & Above
Hard Disk	: 256 GB

PROJECT IMPLEMENTATION

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File Edit Selection View Go Run ··· dashboard.html - SIEM_Project - Visual Studio Code

Welcome | Console Ninja | dashboard.html | style.css | app.py

templates > dashboard.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>SIEM Dashboard</title>
    <link rel="stylesheet" href="/static/style.css">
</head>

<body>

<div class="container">

    <!-- SIDEBAR -->
    <div class="sidebar">
        <h2>SIEM</h2>

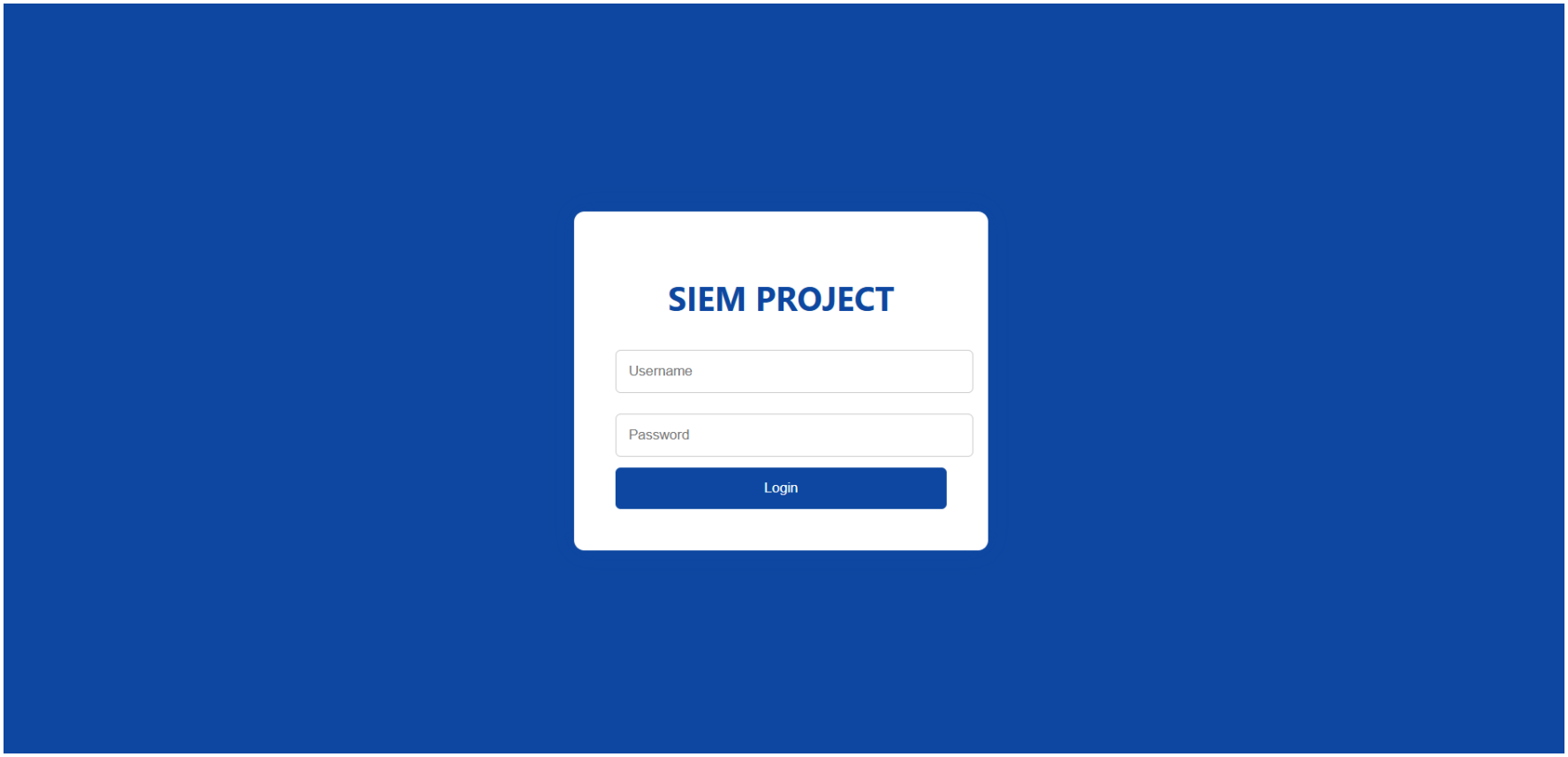
        <button onclick="window.location.href='/'">Home</button>
        <button onclick="window.location.href='/wazuh'">Dashboard</button>
```

PROBLEMS OUTPUT TERMINAL PORTS python

```
PS C:\Users\bprat\OneDrive\Desktop\SIEM_Project> python app.py
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 905-475-721
127.0.0.1 - - [02/Apr/2026 12:53:34] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Apr/2026 12:53:35] "GET /static/style.css HTTP/1.1" 304 -
127.0.0.1 - - [02/Apr/2026 12:53:35] "GET /static/cyber.png HTTP/1.1" 304 -
127.0.0.1 - - [02/Apr/2026 12:53:36] "GET /favicon.ico HTTP/1.1" 404 -
```

0 0 | Screen Reader Optimized | Ln 50, Col 8 | Spaces: 4 | UTF-8 | CRLF | { } HTML | Finish Setup | Go Live | Ninja

PROJECT IMPLEMENTATION



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PROJECT IMPLEMENTATION

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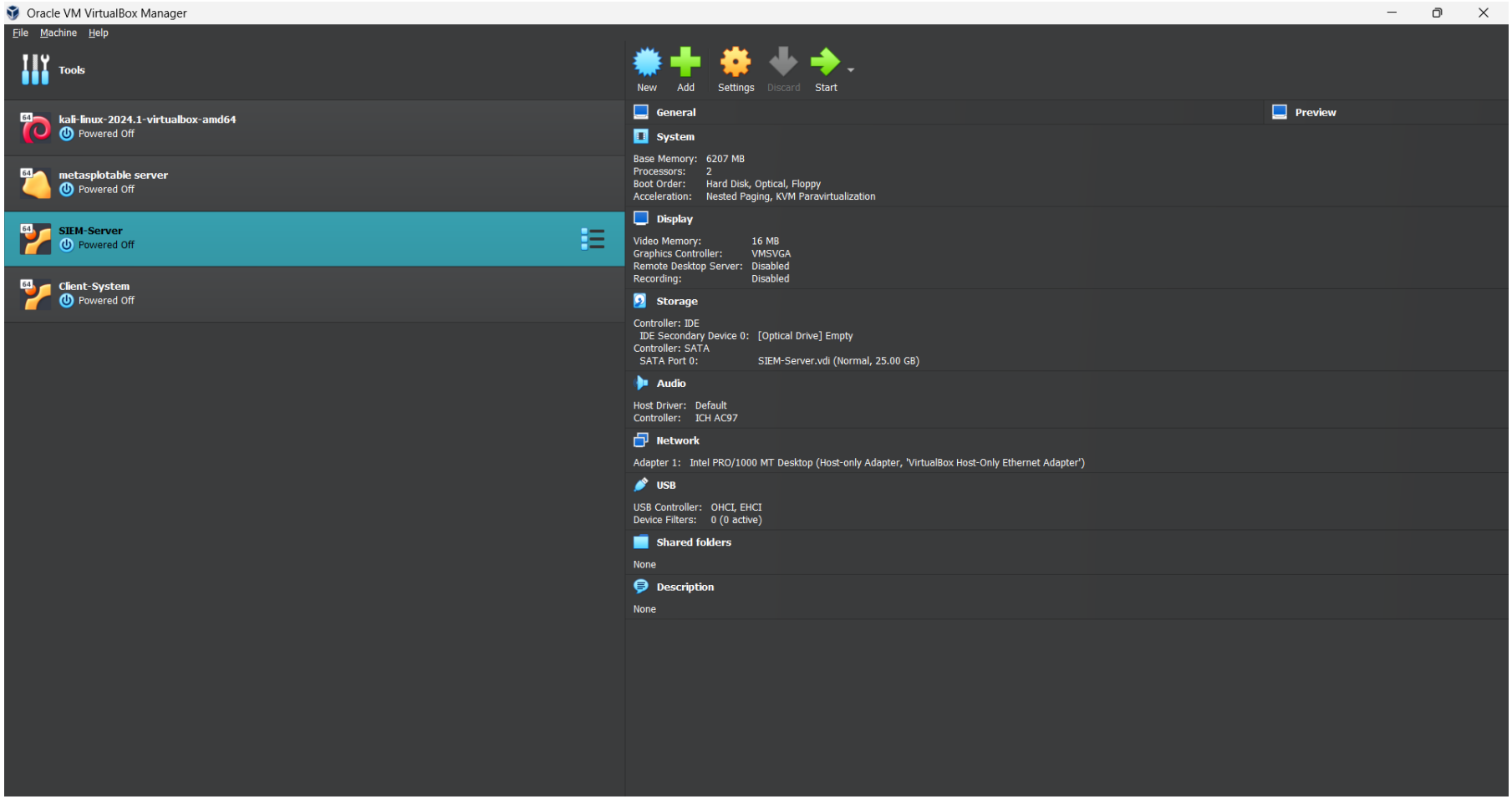

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```
Ubuntu 24.04.4 LTS siem tty1

siem login: chaitu
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Mar 10 03:45:30 PM UTC 2026

  System load: 1.88               Memory usage: 48%   Processes:       125
  Usage of /:  77.8% of 11.21GB   Swap usage:   0%    Users logged in: 0


Expanded Security Maintenance for Applications is not enabled.

5 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


chaitu@siem:~$ _
```

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```
  Usage of /:  77.8% of 11.21GB   Swap usage:   0%    Users logged in: 0


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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


chaitu@siem:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b5:07:e4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s3
       valid_lft 527sec preferred_lft 527sec
    inet6 fe80::a00:27ff:feb5:7e4/64 scope link
       valid_lft forever preferred_lft forever
chaitu@siem:~$ sudo systemctl status wazuh-indexer
[sudo] password for chaitu:
● wazuh-indexer.service - Wazuh-indexer
     Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-03-11 15:19:05 UTC; 16s ago
       Docs: https://documentation.wazuh.com
   Main PID: 738 (java)
      Tasks: 72 (limit: 7086)
     Memory: 2.3G (peak: 2.3G)
        CPU: 1min 35.086s
     CGroup: /system.slice/wazuh-indexer.service
             └─738 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dopensearch.networkaddress.cache.negative.ttl=1>

Mar 11 15:17:07 siem systemd[1]: Starting wazuh-indexer.service - Wazuh-indexer...
Mar 11 15:18:01 siem systemd-entrypoint[738]: WARNING: A terminally deprecated method in java.lang.System has been called
Mar 11 15:18:01 siem systemd-entrypoint[738]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.OpenSearch (file:/usr/share/wazuh>
Mar 11 15:18:01 siem systemd-entrypoint[738]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.OpenSearch
Mar 11 15:18:01 siem systemd-entrypoint[738]: WARNING: System::setSecurityManager will be removed in a future release
Mar 11 15:18:06 siem systemd-entrypoint[738]: WARNING: A terminally deprecated method in java.lang.System has been called
Mar 11 15:18:06 siem systemd-entrypoint[738]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.Security (file:/usr/share/wazuh-i>
Mar 11 15:18:06 siem systemd-entrypoint[738]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.Security
Mar 11 15:18:06 siem systemd-entrypoint[738]: WARNING: System::setSecurityManager will be removed in a future release
Mar 11 15:19:05 siem systemd[1]: Started wazuh-indexer.service - Wazuh-indexer.
lines 1-21/21 (END)
chaitu@siem:~$
```

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```
    Process: 739 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=killed, signal=TERM)
        CPU: 5.147s

Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd:router: INFO: Loaded router module.
Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Mar 11 15:17:51 siem env[1217]: 2026/03/11 15:17:51 wazuh-modulesd:inventory-harvester: INFO: Loaded Inventory harvester module.
chaitu@siem:~$ sudo systemctl start wazuh-manager
chaitu@siem:~$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
     Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-03-11 15:20:32 UTC; 9s ago
    Process: 1387 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
      Tasks: 171 (limit: 7086)
     Memory: 388.1M (peak: 390.3M)
        CPU: 26.325s
     CGroup: /system.slice/wazuh-manager.service
             ├─1453 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1454 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1455 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1458 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1463 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1505 /var/ossec/bin/wazuh-authd
             ├─1523 /var/ossec/bin/wazuh-db
             ├─1562 /var/ossec/bin/wazuh-execd
             ├─1583 /var/ossec/bin/wazuh-analysisd
             ├─1599 /var/ossec/bin/wazuh-syscheckd
             ├─1621 /var/ossec/bin/wazuh-remoted
             ├─1689 /var/ossec/bin/wazuh-monitord
             ├─1701 /var/ossec/bin/wazuh-modulesd
             └─2542 find /var/log -type f -ls

Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd:router: INFO: Loaded router module.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd:inventory-harvester: INFO: Loaded Inventory harvester module.
Mar 11 15:20:30 siem env[1387]: Started wazuh-modulesd...
Mar 11 15:20:32 siem env[1387]: Completed.
Mar 11 15:20:32 siem systemd[1]: Started wazuh-manager.service - Wazuh manager.
chaitu@siem:~$ ^C
chaitu@siem:~$ _
```

PROJECT IMPLEMENTATION

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```
     CGroup: /system.slice/wazuh-manager.service
             ├─1453 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1454 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1455 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─1458 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
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Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd: WARNING: (1230): Invalid element in the configuration: 'provider'.
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Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd:router: INFO: Loaded router module.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Mar 11 15:20:29 siem env[1699]: 2026/03/11 15:20:29 wazuh-modulesd:inventory-harvester: INFO: Loaded Inventory harvester module.
Mar 11 15:20:30 siem env[1387]: Started wazuh-modulesd...
Mar 11 15:20:32 siem env[1387]: Completed.
Mar 11 15:20:32 siem systemd[1]: Started wazuh-manager.service - Wazuh manager.
chaitu@siem:~$ ^C
chaitu@siem:~$ sudo systemctl status wazuh-dashboard
● wazuh-dashboard.service - wazuh-dashboard
     Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-03-11 15:17:07 UTC; 3min 57s ago
   Main PID: 737 (node)
      Tasks: 11 (limit: 7086)
     Memory: 274.0M (peak: 356.2M)
        CPU: 24.485s
     CGroup: /system.slice/wazuh-dashboard.service
             └─737 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size=65536 --unhandled-rejections=warn /usr/share/wazuh-dashboard/s>

Mar 11 15:19:11 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:19:11Z","tags":["error","opensearch","data"],"pid":737,"message":"[R>
Mar 11 15:19:14 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:19:14Z","tags":["error","opensearch","data"],"pid":737,"message":"[R>
Mar 11 15:19:16 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:19:16Z","tags":["error","opensearch","data"],"pid":737,"message":"[R>
Mar 11 15:19:19 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:19:19Z","tags":["error","opensearch","data"],"pid":737,"message":"[R>
Mar 11 15:19:22 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:19:22Z","tags":["info","savedobjects-service"],"pid":737,"message":">
Mar 11 15:19:22 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:19:22Z","tags":["info","plugins-system"],"pid":737,"message":"Starti>
Mar 11 15:19:25 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:19:25Z","tags":["listening","info"],"pid":737,"message":"Server runn>
Mar 11 15:19:25 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:19:25Z","tags":["info","http","server","OpenSearchDashboards"],"pid">
Mar 11 15:20:00 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:20:00Z","tags":["error","plugins","wazuh","cron-scheduler"],"pid":73>
Mar 11 15:20:00 siem opensearch-dashboards[737]: {"type":"log","@timestamp":"2026-03-11T15:20:00Z","tags":["error","plugins","wazuh","cron-scheduler"],"pid":73>

chaitu@siem:~$
```

PROJECT IMPLEMENTATION

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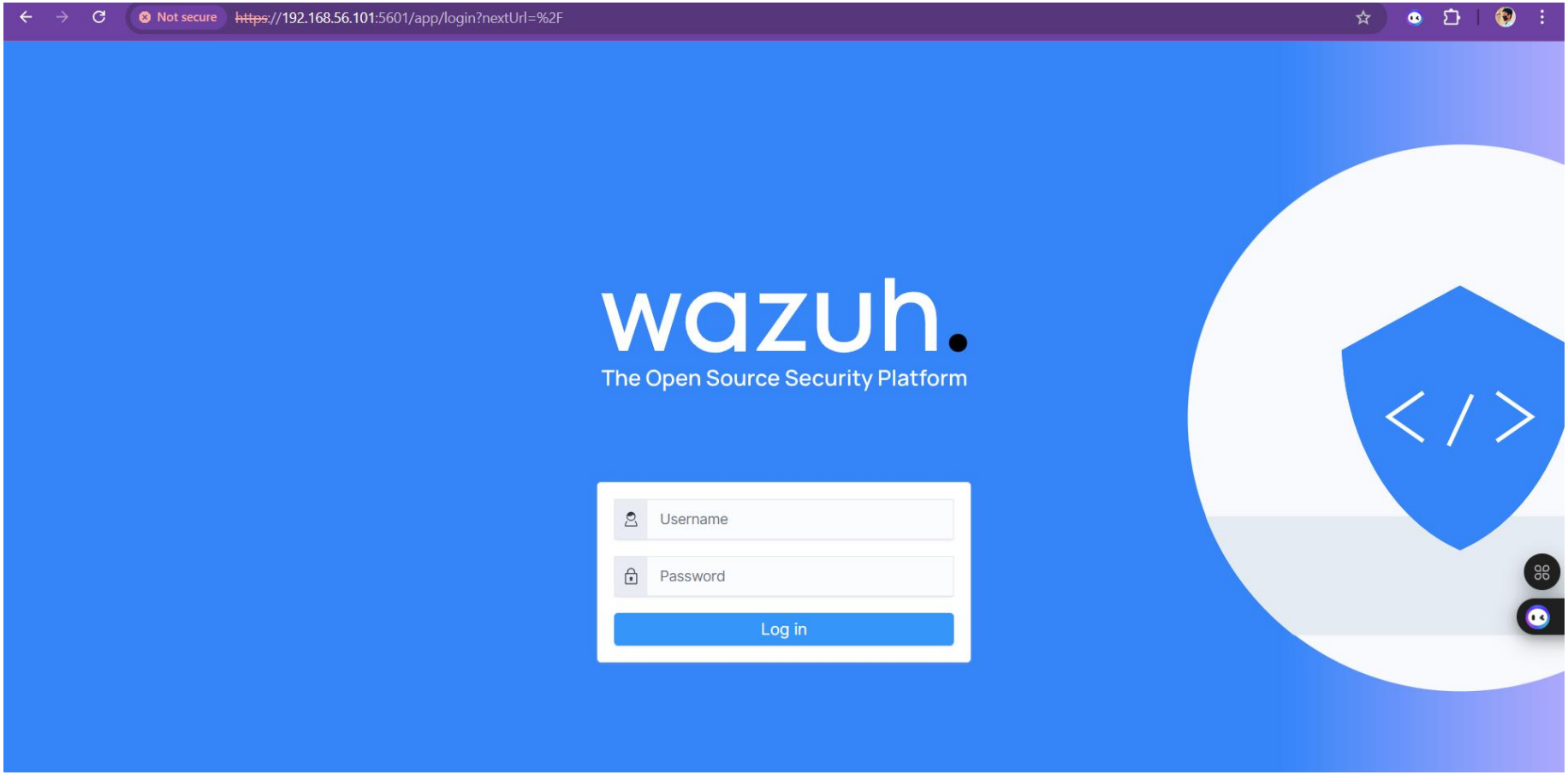
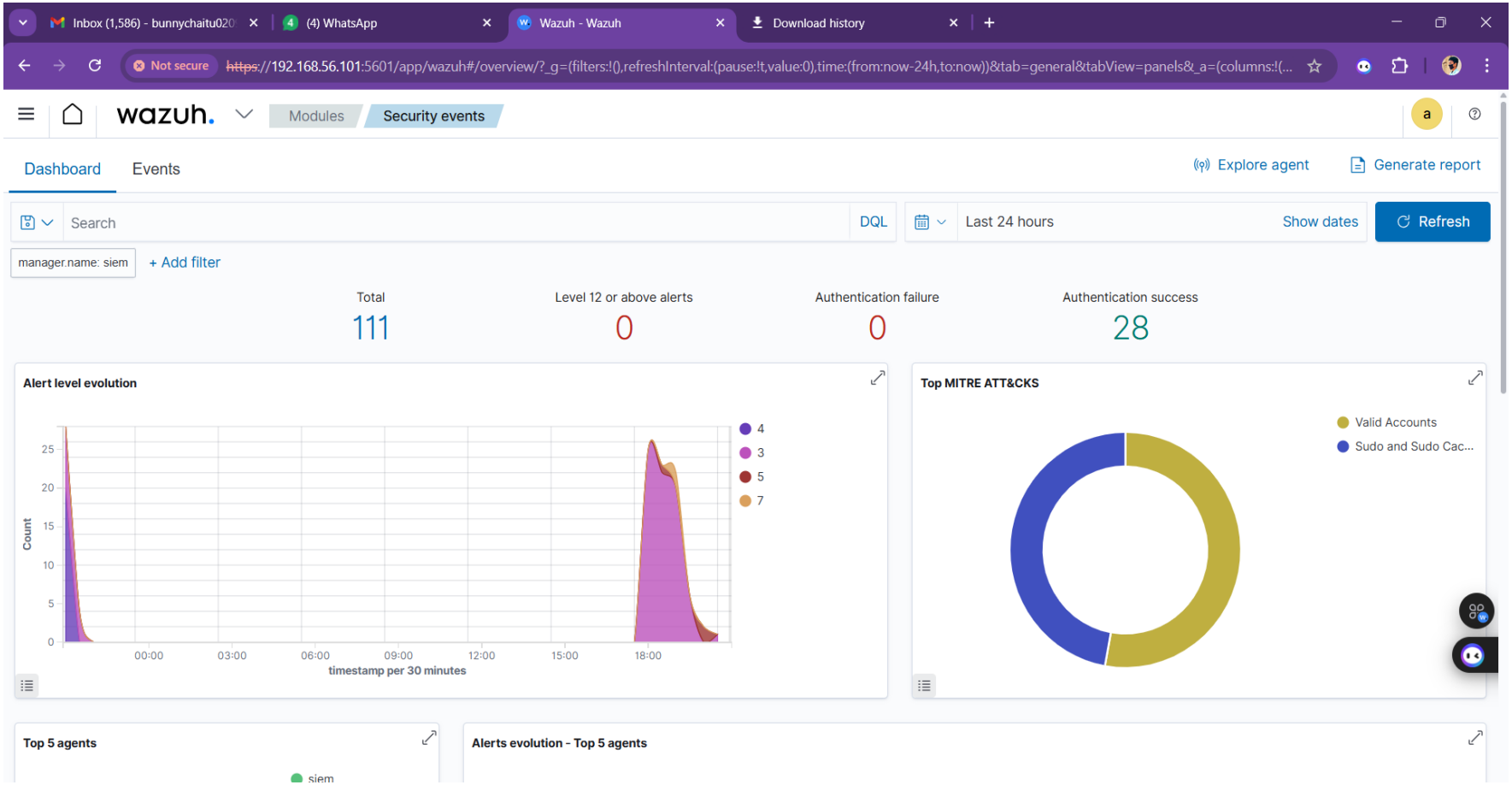


PROJECT IMPLEMENTATION

SVCE TIRUPATI
EDUCATION FOR A BETTER SOCIETY



TESTING

Unit Testing

- Unit testing is testing the smallest testable unit of an application. It is done during the coding phase by the developers.
- Unit Testing helps in finding bugs early in the development process.
- Testing the prediction system code.

Test Cases:

Function	Test Description	Input	Expected Output	Pass Criteria
Log Collection	Collect logs from server	Server log data	Logs successfully stored	Logs successfully received and stored in databas
Log Processing	Test log parsing and normalization	Raw log files	Logs converted into structured format	Logs correctly formatted and stored
Event Detection	Detect suspicious login attempts	Multiple failed login attempts	Security event detected	System correctly identifies suspicious activity
Alert Generation	Trigger alert when a security rule is activated	Security event detected	Alert notification generated	Alert displayed in dashboard and notification sent
Database Storage	Store processed logs in database	Processed log data	Logs stored in centralized database	Logs successfully saved and retrievable

Integration Testing

- Integration Testing is a type of software testing in which the different units, modules or components of a software application are tested as a combined entity.
- It helps in identifying integration issues between the modules and also ensures that the integrated modules work properly before moving to the system testing of the complete application.

Test Case:

Test Scenario	Input	Expected Output	Pass Criteria
Security Event Detection Workflow	Multiple failed login attempts from a user	System detects suspicious activity and generates alert	Alert appears in dashboard and stored in database
Log Monitoring Process	Server and firewall logs	Logs collected and displayed in monitoring dashboard	Logs visible in real-time dashboard

System Testing

- Tests the entire SIEM system after integration
- Verifies log collection from multiple sources
- Checks threat detection and alert generation
- Ensures dashboard displays real-time security events
- Confirms report generation and system functionality

RESULTS

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```
Ubuntu 24.04.4 LTS client tty1

client login: bunny
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Mar  6 11:42:48 AM UTC 2026

  System load:             0.99
  Usage of /:              41.4% of 11.21GB
  Memory usage:            9%
  Swap usage:              0%
  Processes:               103
  Users logged in:         0
  IPv6 address for enp0s3: 2409:40f0:541e:979a:a00:27ff:fe5f:64f4


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


bunny@client:~$ _
```

RESULTS

SVCE TIRUPATI
EDUCATION FOR A BETTER SOCIETY

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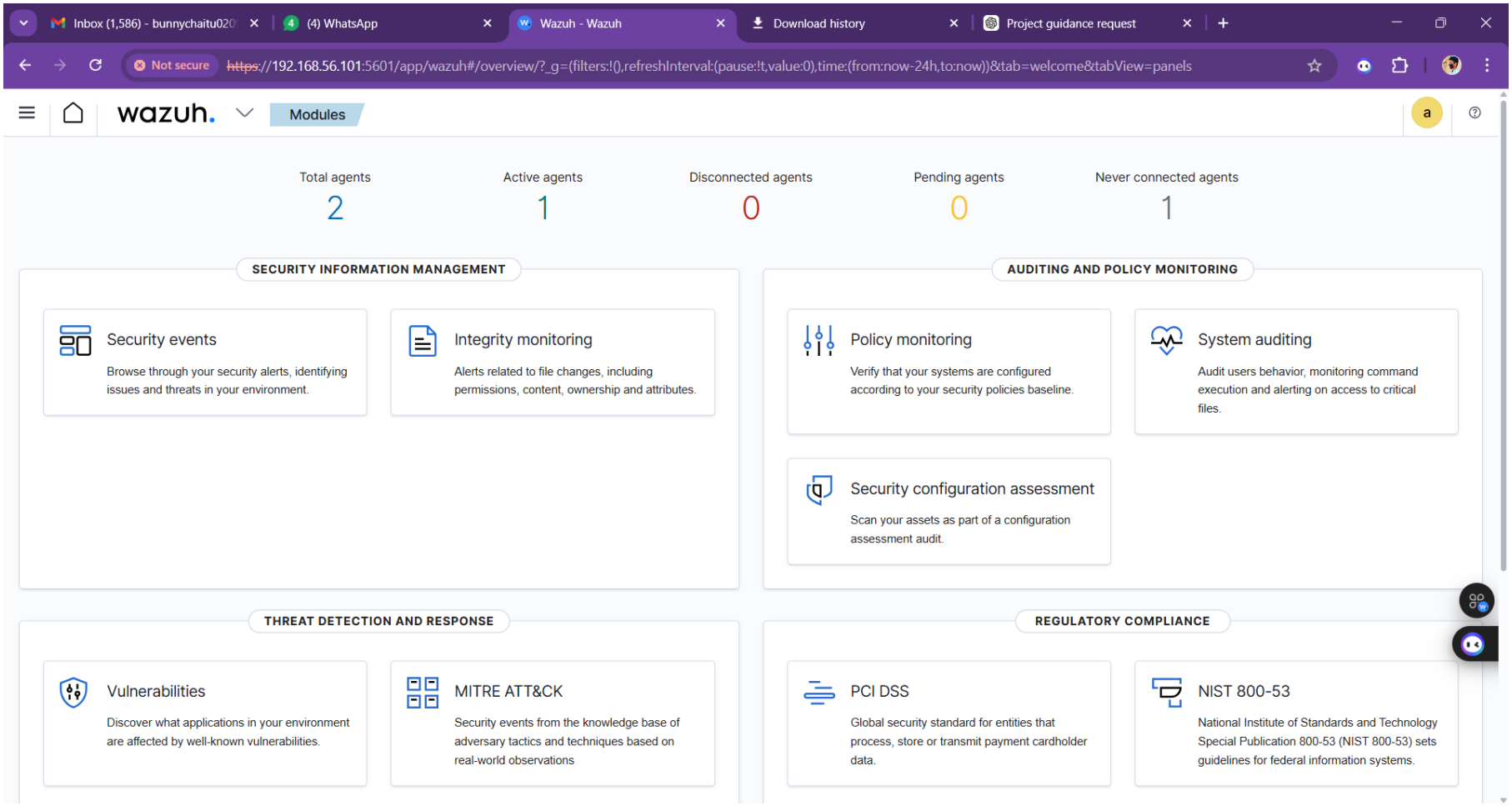
bunny@client:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=6.85 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=2.34 ms
^C
--- 192.168.56.101 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 2.340/4.597/6.854/2.257 ms
bunny@client:~$ _
```

RESULTS

SVCE TIRUPATI
EDUCATION FOR A BETTER SOCIETY



RESULTS

SVCE TIRUPATI
EDUCATION FOR A BETTER SOCIETY

```
bunny@client:~$ sudo systemctl start wazuh-agent
[sudo] password for bunny:
bunny@client:~$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
     Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-03-11 15:53:04 UTC; 1min 22s ago
    Process: 673 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
      Tasks: 27 (limit: 5392)
     Memory: 56.0M (peak: 59.9M)
        CPU: 3.916s
     CGroup: /system.slice/wazuh-agent.service
             ├─818 /var/ossec/bin/wazuh-execd
             ├─827 /var/ossec/bin/wazuh-agentd
             ├─852 /var/ossec/bin/wazuh-syscheckd
             ├─878 /var/ossec/bin/wazuh-logcollector
             └─900 /var/ossec/bin/wazuh-modulesd

Mar 11 15:52:57 client env[673]: Deleting PID file '/var/ossec/var/run/wazuh-syscheckd-912.pid' not used...
Mar 11 15:52:58 client env[673]: Deleting PID file '/var/ossec/var/run/wazuh-agentd-888.pid' not used...
Mar 11 15:52:58 client env[673]: Deleting PID file '/var/ossec/var/run/wazuh-execd-855.pid' not used...
Mar 11 15:52:58 client env[673]: Started wazuh-execd...
Mar 11 15:52:59 client env[673]: Started wazuh-agentd...
Mar 11 15:53:00 client env[673]: Started wazuh-syscheckd...
Mar 11 15:53:01 client env[673]: Started wazuh-logcollector...
Mar 11 15:53:02 client env[673]: Started wazuh-modulesd...
Mar 11 15:53:04 client env[673]: Completed.
Mar 11 15:53:04 client systemd[1]: Started wazuh-agent.service - Wazuh agent.
bunny@client:~$
```

RESULTS

SVCE TIRUPATI
EDUCATION FOR A BETTER SOCIETY

```
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PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=6.85 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=2.34 ms
^C
--- 192.168.56.101 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 2.340/4.597/6.854/2.257 ms
bunny@client:~$ ssh wronguser@localhost
ssh: connect to host localhost port 22: Connection refused
bunny@client:~$ sudo nano /etc/passwd
[sudo] password for bunny:
Sorry, try again.
[sudo] password for bunny:
Sorry, try again.
[sudo] password for bunny:
sudo: 3 incorrect password attempts
bunny@client:~$ _
```

RESULTS

SVCE TIRUPATI
EDUCATION FOR A BETTER SOCIETY

Inbox (1,586) - bunnychaitu020 | (4) WhatsApp | Wazuh - Wazuh | Download history | Project guidance request

Not secure https://192.168.56.101:5601/app/wazuh#/overview/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now))&tab=general&tabView=panels&_a=(columns:!(...

wazuh. Modules Security events

Security Alerts

Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 11, 2026 @ 21:02:54.721	002	client	T1548.003	Privilege Escalation, Defense Evasion	Three failed attempts to run sudo	10	5404
Mar 11, 2026 @ 21:02:34.711	002	client	T1110.001	Credential Access	PAM: User login failed.	5	5503
Mar 11, 2026 @ 20:56:46.282	002	client	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	PAM: Login session opened.	3	5501
Mar 11, 2026 @ 20:56:44.207	002	client	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	PAM: Login session opened.	3	5501
Mar 11, 2026 @ 20:56:40.203	002	client			PAM misconfiguration.	4	5553
Mar 11, 2026 @ 20:56:21.709	002	client			Host-based anomaly detection event (rootcheck).	7	510
Mar 11, 2026 @ 20:56:21.645	002	client			Host-based anomaly detection event (rootcheck).	7	510
Mar 11, 2026 @ 20:56:18.944	002	client			Wazuh agent started.	3	503
Mar 11, 2026 @ 20:50:39.082	000	siem			Wazuh server started.	3	502

FUTURE ENHANCEMENTS

- **Automated Incident Response:** The system can be enhanced to automatically respond to security incidents by blocking malicious IP addresses, disabling compromised user accounts, or isolating infected systems

- **Threat Intelligence Integration:** External threat intelligence feeds can be integrated into the system to identify known malicious IP addresses, domains, and attack signatures.

- **Advanced Visualization and Reporting:** Future improvements can include advanced dashboards and graphical reports that help administrators better analyze security trends and incidents.

- **Scalability Improvements:** The system can be enhanced to handle large volumes of logs generated by large-scale enterprise networks.

CONCLUSIONS

The project successfully demonstrates the design and implementation of a **Security Information and Event Management (SIEM) system** for monitoring security events within a business organization. The system collects logs from various sources such as servers, network devices, and endpoints, and analyzes them to detect suspicious activities and potential security threats. By providing centralized log management, real-time monitoring, and alert generation, the SIEM system helps administrators quickly identify and respond to security incidents. Overall, this project highlights the importance of SIEM solutions in strengthening an organization's cybersecurity infrastructure and improving its ability to detect and manage security threats effectively.

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Thank You